

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the canonical TE Curve Verification Pipeline offline comparison between RCIM Model-Bank Reproduction , recovered original, retuned paper-reference model banks, and repository-owned Wave 1 and Wave 2.1 model candidates. It starts from the current direction-aware comparison matrix.

Dataset And Split

- dataset config: config/datasets/transmission_error_dataset.yaml ;
- dataset root: data\simplified_dataset ;
- comparison mode: full_directional_candidate_matrix ;
- candidate count: 231 ;
- held-out curve count before candidate filtering: 194 ;
- percentage-error denominator: peak_to_peak_truth ;
- Fw candidates are evaluated only on forward curves;
- Bw candidates are evaluated only on backward curves;
- global candidates are evaluated on both directions and reported with direction-separated metrics.

Candidate Inventory

Candidate	Family	Source	Kind	Surface	Valid Directions
SVM19_Fw	SVM	rcim_track1	track1_reference_bank	Fw	forward
SVM19_Bw	SVM	rcim_track1	track1_reference_bank	Bw	backward
MLP19_Fw	MLP	rcim_track1	track1_reference_bank	Fw	forward
MLP19_Bw	MLP	rcim_track1	track1_reference_bank	Bw	backward
RF19_Fw	RF	rcim_track1	track1_reference_bank	Fw	forward
RF19_Bw	RF	rcim_track1	track1_reference_bank	Bw	backward
DT19_Fw	DT	rcim_track1	track1_reference_bank	Fw	forward
DT19_Bw	DT	rcim_track1	track1_reference_bank	Bw	backward

Candidate	Family	Source	Kind	Surface	Valid Directions
ET19_Fw	ET	rcim_track1	track1_reference_bank	Fw	forward
ET19_Bw	ET	rcim_track1	track1_reference_bank	Bw	backward
ERT19_Fw	ERT	rcim_track1	track1_reference_bank	Fw	forward
ERT19_Bw	ERT	rcim_track1	track1_reference_bank	Bw	backward
GBM19_Fw	GBM	rcim_track1	track1_reference_bank	Fw	forward
GBM19_Bw	GBM	rcim_track1	track1_reference_bank	Bw	backward
HGBM19_Fw	HGBM	rcim_track1	track1_reference_bank	Fw	forward
HGBM19_Bw	HGBM	rcim_track1	track1_reference_bank	Bw	backward
XGBM19_Fw	XGBM	rcim_track1	track1_reference_bank	Fw	forward
XGBM19_Bw	XGBM	rcim_track1	track1_reference_bank	Bw	backward
LGBM19_Fw	LGBM	rcim_track1	track1_reference_bank	Fw	forward
LGBM19_Bw	LGBM	rcim_track1	track1_reference_bank	Bw	backward
ELM19_Fw	ELM	rcim_track1	track1_reference_bank	Fw	forward
ELM19_Bw	ELM	rcim_track1	track1_reference_bank	Bw	backward
rcim_original_SVM19_Fw	SVM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_MLP19_Fw	MLP	rcim_original	track1_reference_bank	Fw	forward
rcim_original_RF19_Fw	RF	rcim_original	track1_reference_bank	Fw	forward
rcim_original_DT19_Fw	DT	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ET19_Fw	ET	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ERT19_Fw	ERT	rcim_original	track1_reference_bank	Fw	forward
rcim_original_GBM19_Fw	GBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_HGBM19_Fw	HGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_XGBM19_Fw	XGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_LGBM19_Fw	LGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ELM19_Fw	ELM	rcim_original	track1_reference_bank	Fw	forward
rcim_retuned_SVM19_Fw	SVM	rcim_retuned	track1_reference_bank	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
rcim_retuned_MLP19_Fw	MLP	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_RF19_Fw	RF	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_DT19_Fw	DT	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ET19_Fw	ET	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ERT19_Fw	ERT	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_GBM19_Fw	GBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_HGBM19_Fw	HGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_XGBM19_Fw	XGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_LGBM19_Fw	LGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ELM19_Fw	ELM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_SVM19_Bw	SVM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_MLP19_Bw	MLP	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_RF19_Bw	RF	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_DT19_Bw	DT	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ET19_Bw	ET	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ERT19_Bw	ERT	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_GBM19_Bw	GBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_HGBM19_Bw	HGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_XGBM19_Bw	XGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_LGBM19_Bw	LGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ELM19_Bw	ELM	rcim_retuned	track1_reference_bank	Bw	backward
temporal_convolution_global	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
temporal_convolution_Fw	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward
temporal_convolution_Bw	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
gru_sequence_global	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
gru_sequence_Fw	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
gru_sequence_Bw	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
lstm_sequence_global	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
lstm_sequence_Fw	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward
lstm_sequence_Bw	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
feedforward_global	feedforward	wave1	wave1_exported_model	global	forward, backward
feedforward_Fw	feedforward	wave1	wave1_exported_model	Fw	forward
feedforward_Bw	feedforward	wave1	wave1_exported_model	Bw	backward
harmonic_regression_global	harmonic_regression	wave1	wave1_exported_model	global	forward, backward
harmonic_regression_Fw	harmonic_regression	wave1	wave1_exported_model	Fw	forward
harmonic_regression_Bw	harmonic_regression	wave1	wave1_exported_model	Bw	backward
periodic_mlp_global	periodic_mlp	wave1	wave1_exported_model	global	forward, backward
periodic_mlp_Fw	periodic_mlp	wave1	wave1_exported_model	Fw	forward
periodic_mlp_Bw	periodic_mlp	wave1	wave1_exported_model	Bw	backward
residual_harmonic_mlp_global	residual_harmonic_mlp	wave1	wave1_exported_model	global	forward, backward
residual_harmonic_mlp_Fw	residual_harmonic_mlp	wave1	wave1_exported_model	Fw	forward
residual_harmonic_mlp_Bw	residual_harmonic_mlp	wave1	wave1_exported_model	Bw	backward
tree_global	tree	wave1	wave1_exported_model	global	forward, backward
tree_Fw	tree	wave1	wave1_exported_model	Fw	forward
tree_Bw	tree	wave1	wave1_exported_model	Bw	backward
paper_original_best_Fw	best_composite	rcim_original	composite_reference_bank	Fw	forward
paper_retuned_best_Fw	best_composite	rcim_retuned	composite_reference_bank	Fw	forward
track1_best_Fw	best_composite	rcim_track1	composite_reference_bank	Fw	forward
paper_retuned_best_Bw	best_composite	rcim_retuned	composite_reference_bank	Bw	backward
track1_best_Bw	best_composite	rcim_track1	composite_reference_bank	Bw	backward
periodic_temporal_convolution_global	periodic_temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
periodic_temporal_convolution_Fw	periodic_temporal_convolution	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_temporal_convolution_Bw	periodic_temporal_convolution	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
periodic_gru_sequence_global	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	global	forward, backward
periodic_gru_sequence_Fw	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_gru_sequence_Bw	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
periodic_lstm_sequence_global	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	global	forward, backward
periodic_lstm_sequence_Fw	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_lstm_sequence_Bw	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_sparse_rcim_Fw	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_sparse_rcim_Bw	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_dense240_Fw	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_dense240_Bw	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_dense360_Fw	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_dense360_Bw	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
residual_harmonic_lstm_sequence_sparse_rcim_Fw	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_sparse_rcim_Bw	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_lstm_sequence_dense240_Fw	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_dense240_Bw	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_lstm_sequence_dense360_Fw	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_dense360_Bw	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
polished_rcim_model_bank_reproduction_SVM19_Fw	SVM	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_MLP19_Fw	MLP	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_RF19_Fw	RF	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_DT19_Fw	DT	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_ET19_Fw	ET	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_ERT19_Fw	ERT	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_GBM19_Fw	GBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_HGBM19_Fw	HGBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
reproduction_HGBM19_Fw		on			
polished_rcim_model_bank_reproduction_XGBM19_Fw	XGBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_LGBM19_Fw	LGBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
sequential_residual_offset_probe_global	sequential_residual_offset_probe	track2f_offset_aware_probe_registry	wave1_registry_model	global	forward, backward
sequential_residual_offset_probe_Fw	sequential_residual_offset_probe	track2f_offset_aware_probe_registry	wave1_registry_model	Fw	forward
track2f_bis_clean_sequential_residual_offset_global	track2f_bis_clean_sequential_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	global	forward, backward
track2f_bis_clean_sequential_residual_offset_Fw	track2f_bis_clean_sequential_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	Fw	forward
track2f_bis_harmonic_residual_offset_global	track2f_bis_harmonic_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	global	forward, backward
track2f_bis_harmonic_residual_offset_Fw	track2f_bis_harmonic_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	Fw	forward
track2g_curve_aware_pointwise_control_global	track2g_curve_aware_pointwise_control	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_pointwise_control_Fw	track2g_curve_aware_pointwise_control	track2g_curve_aware_training_registry	wave1_registry_model	Fw	forward
track2g_curve_aware_raw_centered_shape_global	track2g_curve_aware_raw_centered_shape	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_raw_centered_shape_Fw	track2g_curve_aware_raw_centered_shape	track2g_curve_aware_training_registry	wave1_registry_model	Fw	forward
track2g_curve_aware_raw_offset_global	track2g_curve_aware_raw_offset	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_raw_offset_Fw	track2g_curve_aware_raw_offset	track2g_curve_aware_training_registry	wave1_registry_model	Fw	forward
track2g_curve_aware_full_curve_composite_global	track2g_curve_aware_full_curve_composite	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_full_curve_composite_Fw	track2g_curve_aware_full_curve_composite	track2g_curve_aware_training_registry	wave1_registry_model	Fw	forward
track2h_mae_robust_global	track2h_mae_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_mae_robust_Fw	track2h_mae_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
egistry					
track2h_smooth_l1_robust_global	track2h_smooth_l1_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_smooth_l1_robust_Fw	track2h_smooth_l1_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	Fw	forward
track2h_log_cosh_robust_global	track2h_log_cosh_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_log_cosh_robust_Fw	track2h_log_cosh_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	Fw	forward
track2h_quantile_p10_p50_p90_global	track2h_quantile_p10_p50_p90	track2h_quantile_probabilistic_registry	wave1_registry_model	global	forward, backward
track2h_quantile_p10_p50_p90_Fw	track2h_quantile_p10_p50_p90	track2h_quantile_probabilistic_registry	wave1_registry_model	Fw	forward
track2h_gaussian_nll_global	track2h_gaussian_nll	track2h_quantile_probabilistic_registry	wave1_registry_model	global	forward, backward
track2h_gaussian_nll_Fw	track2h_gaussian_nll	track2h_quantile_probabilistic_registry	wave1_registry_model	Fw	forward
track2h_mdn_k2_global	track2h_mdn_k2	track2h_mixture_density_heads_registry	wave1_registry_model	global	forward, backward
track2h_mdn_k2_Fw	track2h_mdn_k2	track2h_mixture_density_heads_registry	wave1_registry_model	Fw	forward
track2h_mdn_k3_global	track2h_mdn_k3	track2h_mixture_density_heads_registry	wave1_registry_model	global	forward, backward
track2h_mdn_k3_Fw	track2h_mdn_k3	track2h_mixture_density_heads_registry	wave1_registry_model	Fw	forward
track2h_l_gru_offset_residual_global	track2h_l_gru_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	global	forward, backward
track2h_l_gru_offset_residual_Fw	track2h_l_gru_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	Fw	forward
track2h_l_causal_tcn_offset_residual_global	track2h_l_causal_tcn_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	global	forward, backward
track2h_l_causal_tcn_offset_residual_Fw	track2h_l_causal_tcn_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	Fw	forward
wave3_harmonic_prior_residual_pointwise_control_global	wave3_harmonic_prior_residual_pointwise_control	wave3_harmonic_prior_residual_registry	wave1_registry_model	global	forward, backward
wave3_harmonic_prior_residual_pointwise_control_Fw	wave3_harmonic_prior_residual_pointwise_control	wave3_harmonic_prior_residual_registry	wave1_registry_model	Fw	forward
wave3_harmonic_prior_residual_	wave3_harmonic_prior_residual_smooth	wave3_harmonic_prior_residual_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
smooth_l1_structured_global	h_l1_structured	try			
wave3_harmonic_prior_residual_smooth_l1_structured_Fw	wave3_harmonic_prior_residual_smooth_l1_structured	wave3_harmonic_prior_residual_registry	wave1_registry_model	Fw	forward
wave52b_offset_centered_shape_harmonic_global	wave52b_offset_harmonic_guided	wave52b_offset_harmonic_guided_registry	wave1_registry_model	global	forward, backward
wave52b_offset_centered_shape_harmonic_Fw	wave52b_offset_harmonic_guided	wave52b_offset_harmonic_guided_registry	wave1_registry_model	Fw	forward
polished_feedforward_global	feedforward	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_feedforward_Fw	feedforward	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_harmonic_regression_global	harmonic_regression	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_harmonic_regression_Fw	harmonic_regression	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_periodic_mlp_global	periodic_mlp	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_mlp_Fw	periodic_mlp	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_residual_harmonic_mlp_global	residual_harmonic_mlp	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_mlp_Fw	residual_harmonic_mlp	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_tree_global	tree	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_tree_Fw	tree	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_periodic_mlp_harmonic_global	periodic_mlp_harmonic	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_mlp_harmonic_Fw	periodic_mlp_harmonic	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_temporal_convolution_global	temporal_convolution	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_temporal_convolution_Fw	temporal_convolution	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_gru_sequence_global	gru_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_gru_sequence_Fw	gru_sequence	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_lstm_sequence_global	lstm_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_lstm_sequence_Fw	lstm_sequence	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_periodic_temporal_convolution_global	periodic_temporal_convolution	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_temporal_convolution_Fw	periodic_temporal_convolution	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_periodic_gru_sequence_global	periodic_gru_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_gru_sequence_Fw	periodic_gru_sequence	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_periodic_lstm_sequence_global	periodic_lstm_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_lstm_sequence_Fw	periodic_lstm_sequence	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_sparse_rcim_Fw	residual_harmonic_gru_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_dense240_Fw	residual_harmonic_gru_sequence_dense240	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_dense360_Fw	residual_harmonic_gru_sequence_dense360	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_sparse_rcim_Fw	residual_harmonic_lstm_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
lstm_sequence_sparse_rcim_Fw					
polished_residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_dense240_Fw	residual_harmonic_lstm_sequence_dense240	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_dense360_Fw	residual_harmonic_lstm_sequence_dense360	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave3_1_sequential_residual_offset_probe_global	wave3_1_sequential_residual_offset_probe	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_1_sequential_residual_offset_probe_Fw	wave3_1_sequential_residual_offset_probe	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave3_2_clean_sequential_residual_offset_global	wave3_2_clean_sequential_residual_offset	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_2_clean_sequential_residual_offset_Fw	wave3_2_clean_sequential_residual_offset	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave3_2_harmonic_residual_offset_global	wave3_2_harmonic_residual_offset	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_2_harmonic_residual_offset_Fw	wave3_2_harmonic_residual_offset	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave3_3_curve_aware_pointwise_control_global	wave3_3_curve_aware_pointwise_control	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_curve_aware_pointwise_control_Fw	wave3_3_curve_aware_pointwise_control	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave3_3_raw_centered_shape_curve_aware_global	wave3_3_raw_centered_shape_curve_aware	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_raw_centered_shape_curve_aware_Fw	wave3_3_raw_centered_shape_curve_aware	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave3_3_raw_offset_curve_aware_global	wave3_3_raw_offset_curve_aware	polished_model_development_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_wave3_3_raw_offset_curve_aware_Fw	wave3_3_raw_offset_curve_aware	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave3_3_full_curve_composite_global	wave3_3_full_curve_composite	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_full_curve_composite_Fw	wave3_3_full_curve_composite	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_1_mae_robust_loss_global	wave4_1_mae_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_mae_robust_loss_Fw	wave4_1_mae_robust_loss	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_1_smooth_l1_robust_loss_global	wave4_1_smooth_l1_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_smooth_l1_robust_loss_Fw	wave4_1_smooth_l1_robust_loss	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_1_log_cosh_robust_loss_global	wave4_1_log_cosh_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_log_cosh_robust_loss_Fw	wave4_1_log_cosh_robust_loss	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_2_quantile_p10_p50_p90_global	wave4_2_quantile_p10_p50_p90	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_2_quantile_p10_p50_p90_Fw	wave4_2_quantile_p10_p50_p90	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_2_gaussian_nll_global	wave4_2_gaussian_nll	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_2_gaussian_nll_Fw	wave4_2_gaussian_nll	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_3_mixture_density_k2_global	wave4_3_mixture_density_k2	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_3_mixture_density_k2_Fw	wave4_3_mixture_density_k2	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_3_mixture_density_k3_global	wave4_3_mixture_density_k3	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_3_mixture_density_k3_Fw	wave4_3_mixture_density_k3	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_4_gru_latent_offset_residual_global	wave4_4_gru_latent_offset_residual	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_4_gru_latent_offset_residual_Fw	wave4_4_gru_latent_offset_residual	polished_model_development_registry	wave1_registry_model	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_wave4_4_causal_tcn_latent_offset_residual_global	wave4_4_causal_tcn_latent_offset_residual	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_4_causal_tcn_latent_offset_residual_Fw	wave4_4_causal_tcn_latent_offset_residual	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave5_1_harmonic_prior_pointwise_control_global	wave5_1_harmonic_prior_pointwise_control	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave5_1_harmonic_prior_pointwise_control_Fw	wave5_1_harmonic_prior_pointwise_control	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	wave5_1_harmonic_prior_smooth_l1_structured	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave5_1_harmonic_prior_smooth_l1_structured_Fw	wave5_1_harmonic_prior_smooth_l1_structured	polished_model_development_registry	wave1_registry_model	Fw	forward

Best Composite Reference Models

These candidates combine the approved best harmonic-wise cells into one TE Curve Verification Pipeline curve-reconstruction candidate. They are also repeated inside the source-group tables below, but this section keeps the composed models explicit.

Candidate	Source	Surface	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
paper_original_best_Fw	rcim_original	Fw	forward	0.002769	0.002951	6.250	13.827
paper_retuned_best_Fw	rcim_retuned	Fw	forward	0.001839	0.002041	4.109	9.866
track1_best_Fw	rcim_track1	Fw	forward	0.003014	0.003204	6.819	11.638
paper_retuned_best_Bw	rcim_retuned	Bw	backward	0.003675	0.004284	7.572	15.645
track1_best_Bw	rcim_track1	Bw	backward	0.005027	0.005212	11.860	48.106

Forward Comparison

Original Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_original_ERT19_Fw	0.001471	0.001677	3.253	6.145
rcim_original_RF19_Fw	0.001767	0.001971	3.940	6.872
rcim_original_LGBM19_Fw	0.001801	0.002004	4.017	10.054
rcim_original_DT19_Fw	0.001919	0.002114	4.306	9.063
rcim_original_GBM19_Fw	0.001921	0.002122	4.312	8.193
rcim_original_HGBM19_Fw	0.002011	0.002217	4.493	10.617
rcim_original_ET19_Fw	0.002232	0.002432	4.985	11.357
rcim_original_XGBM19_Fw	0.002594	0.002814	5.805	10.574
paper_original_best_Fw	0.002769	0.002951	6.250	13.827
rcim_original_SVM19_Fw	0.003052	0.003324	6.767	13.827
rcim_original_ELM19_Fw	0.005423	0.005731	12.130	28.721
rcim_original_MLP19_Fw	0.018754	0.022589	42.943	107.515

Retuned Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912
rcim_retuned_RF19_Fw	0.001487	0.001699	3.292	5.998
rcim_retuned_ERT19_Fw	0.001807	0.002010	4.039	7.599
paper_retuned_best_Fw	0.001839	0.002041	4.109	9.866
rcim_retuned_HGBM19_Fw	0.001851	0.002056	4.126	9.401
rcim_retuned_LGBM19_Fw	0.001851	0.002055	4.135	9.866
rcim_retuned_DT19_Fw	0.001919	0.002114	4.306	9.063
rcim_retuned_ET19_Fw	0.002001	0.002196	4.426	9.533
rcim_retuned_XGBM19_Fw	0.002054	0.002264	4.588	10.488
rcim_retuned_SVM19_Fw	0.003167	0.003428	7.035	12.971
rcim_retuned_ELM19_Fw	0.007182	0.007463	16.181	40.024
rcim_retuned_MLP19_Fw	0.016647	0.020154	38.510	86.197

RCIM Model-Bank Reproduction Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
RF19_Fw	0.002164	0.002371	4.841	11.061
ERT19_Fw	0.002295	0.002491	5.163	13.141
GBM19_Fw	0.002779	0.002981	6.238	12.995
track1_best_Fw	0.003014	0.003204	6.819	11.638
DT19_Fw	0.003122	0.003313	7.011	13.928
SVM19_Fw	0.003236	0.003487	7.185	11.841
HGBM19_Fw	0.003251	0.003464	7.315	13.802
ET19_Fw	0.003267	0.003467	7.339	14.670
XGBM19_Fw	0.004190	0.004396	9.407	22.660
LGBM19_Fw	0.006812	0.007009	15.415	30.398
ELM19_Fw	0.007281	0.007573	16.352	36.326
MLP19_Fw	0.038690	0.047157	89.567	201.437

Wave 1 Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
tree_global	0.002998	0.003364	6.590	11.601
tree_Fw	0.003053	0.003395	6.731	11.995
periodic_mlp_Fw	0.003366	0.003794	7.467	12.904
periodic_mlp_global	0.003380	0.003791	7.493	14.349
harmonic_regression_Fw	0.003394	0.003807	7.534	11.391
residual_harmonic_mlp_Fw	0.003434	0.003876	7.618	13.617
feedforward_Fw	0.003466	0.003904	7.685	13.132
feedforward_global	0.003550	0.003979	7.916	16.195
residual_harmonic_mlp_global	0.003649	0.004051	8.123	14.713
harmonic_regression_global	0.018314	0.018547	41.749	79.404

Wave 2.1 Temporal Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_lstm_sequence_global	0.002726	0.002959	6.142	14.092
periodic_gru_sequence_global	0.002777	0.003023	6.267	13.580
periodic_gru_sequence_Fw	0.003186	0.003438	7.077	11.974
periodic_lstm_sequence_Fw	0.003266	0.003550	7.258	11.961
gru_sequence_Fw	0.003330	0.003762	7.378	13.029
periodic_temporal_convolution_Fw	0.003335	0.003708	7.404	12.518
lstm_sequence_Fw	0.003366	0.003800	7.450	11.807
periodic_temporal_convolution_global	0.003407	0.003724	7.581	14.480
lstm_sequence_global	0.003445	0.003863	7.642	12.032
temporal_convolution_global	0.003508	0.003928	7.792	14.045
gru_sequence_global	0.003546	0.003975	7.869	14.344
temporal_convolution_Fw	0.003603	0.004031	8.028	14.674

Wave 2.3 Residual Harmonic Temporal Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_gru_sequence_sparse_rcim_Fw	0.003194	0.003499	7.083	13.041
residual_harmonic_lstm_sequence_sparse_rcim_Fw	0.003229	0.003533	7.164	12.682
residual_harmonic_lstm_sequence_sparse_rcim_global	0.003245	0.003562	7.192	12.555
residual_harmonic_gru_sequence_sparse_rcim_global	0.003287	0.003611	7.282	12.461
residual_harmonic_lstm_sequence_dense240_global	0.006247	0.008597	14.061	20.066
residual_harmonic_gru_sequence_dense240_global	0.006534	0.008959	14.732	20.898
residual_harmonic_gru_sequence_dense240_Fw	0.006983	0.009275	15.722	19.139
residual_harmonic_lstm_sequence_dense240_Fw	0.007042	0.009370	15.868	19.358
residual_harmonic_lstm_sequence_dense360_Fw	0.007731	0.010235	17.430	21.285
residual_harmonic_gru_sequence_dense360_global	0.007844	0.011211	17.679	22.520
residual_harmonic_gru_sequence_dense360_Fw	0.007869	0.010574	17.740	20.918
residual_harmonic_lstm_sequence_dense360_global	0.008760	0.012992	19.759	24.068

Wave 3.1 Offset-Aware Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
sequential_residual_offset_probe_Fw	0.003377	0.003799	7.487	13.103
sequential_residual_offset_probe_global	0.003425	0.003839	7.599	12.346

Wave 3.2 Harmonic-Offset Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2f_bis_harmonic_residual_offset_Fw	0.002850	0.003108	6.286	12.796
track2f_bis_harmonic_residual_offset_global	0.003255	0.003547	7.224	12.114
track2f_bis_clean_sequential_residual_offset_global	0.003420	0.003841	7.590	13.476
track2f_bis_clean_sequential_residual_offset_Fw	0.003439	0.003870	7.632	12.515

Wave 3.3 Curve-Aware Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_raw_centered_shape_Fw	0.003174	0.003429	7.047	12.981
track2g_curve_aware_full_curve_composite_Fw	0.003251	0.003515	7.209	11.876
track2g_curve_aware_raw_centered_shape_global	0.003248	0.003556	7.218	12.268
track2g_curve_aware_raw_offset_Fw	0.003269	0.003588	7.268	11.943
track2g_curve_aware_raw_offset_global	0.003359	0.003630	7.429	12.737
track2g_curve_aware_pointwise_control_Fw	0.003362	0.003612	7.474	13.966
track2g_curve_aware_pointwise_control_global	0.003499	0.003784	7.771	13.513
track2g_curve_aware_full_curve_composite_global	0.085677	0.085733	188.514	319.710

Wave 5.2 series Robust-Loss Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_mae_robust_Fw	0.003134	0.003382	6.956	12.470
track2h_smooth_l1_robust_global	0.003222	0.003507	7.161	12.021
track2h_mae_robust_global	0.003235	0.003544	7.189	12.856
track2h_smooth_l1_robust_Fw	0.003300	0.003545	7.342	13.401
track2h_log_cosh_robust_Fw	0.003344	0.003595	7.427	12.598
track2h_log_cosh_robust_global	0.003515	0.003825	7.792	14.729

Wave 5.2 series Quantile Probabilistic Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_gaussian_nll_global	0.002951	0.003246	6.524	15.237
track2h_gaussian_nll_Fw	0.003156	0.003415	7.008	10.991
track2h_quantile_p10_p50_p90_global	0.003188	0.003469	7.059	12.765
track2h_quantile_p10_p50_p90_Fw	0.003276	0.003545	7.279	12.393

Wave 5.2 series Mixture Density Heads Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_mdn_k3_Fw	0.003226	0.003487	7.164	11.702
track2h_mdn_k2_global	0.003263	0.003582	7.261	14.354
track2h_mdn_k2_Fw	0.003329	0.003593	7.388	12.771
track2h_mdn_k3_global	0.003415	0.003713	7.594	15.012

Wave 4.4 Latent-State Hysteresis Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_l_causal_tcn_offset_residual_global	0.003267	0.003715	7.241	12.649
track2h_l_causal_tcn_offset_residual_Fw	0.003476	0.003939	7.717	13.330
track2h_l_gru_offset_residual_Fw	0.003549	0.003996	7.873	12.664
track2h_l_gru_offset_residual_global	0.087641	0.087704	192.524	320.620

Wave 5.1 Harmonic-Prior Residual Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave3_harmonic_prior_residual_pointwise_control_global	0.003279	0.003550	7.289	14.109
wave3_harmonic_prior_residual_pointwise_control_Fw	0.003374	0.003655	7.501	12.728
wave3_harmonic_prior_residual_smooth_l1_structured_Fw	0.003514	0.003768	7.812	13.703
wave3_harmonic_prior_residual_smooth_l1_structured_global	0.087837	0.087893	192.823	321.880

Wave 5.2B Offset And Harmonic Guided Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_Fw	0.058653	0.058748	128.839	229.455
wave52b_offset_centered_shape_harmonic_global	0.080306	0.080373	176.296	312.921

Polished Model-Development Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_harmonic_regression_Fw	0.003230	0.003494	7.185	11.606
polished_tree_global	0.066315	0.066379	145.420	257.447
polished_tree_Fw	0.066315	0.066379	145.420	257.447
polished_harmonic_regression_global	0.081589	0.081641	179.273	315.612
polished_wave4_2_gaussian_nll_global	0.082468	0.082526	181.304	319.127
polished_wave4_3_mixture_density_k2_global	0.082547	0.082604	181.584	320.574
polished_wave5_1_harmonic_prior_pointwise_control_Fw	0.082881	0.082943	181.809	315.245
polished_wave4_2_quantile_p10_p50_p90_Fw	0.083077	0.083129	182.617	319.159
polished_wave4_3_mixture_density_k3_Fw	0.083190	0.083242	182.913	318.098
polished_wave4_2_quantile_p10_p50_p90_global	0.083629	0.083685	183.639	320.063
polished_wave3_3_raw_offset_curve_aware_Fw	0.083671	0.083727	184.231	317.654
polished_wave4_2_gaussian_nll_Fw	0.083971	0.084023	184.509	323.590
polished_wave3_3_raw_offset_curve_aware_global	0.084156	0.084216	185.372	318.550
polished_wave4_1_mae_robust_loss_Fw	0.084308	0.084363	185.612	315.815
polished_wave4_1_mae_robust_loss_global	0.084350	0.084403	185.643	318.504
polished_wave4_1_log_cosh_robust_loss_Fw	0.084634	0.084695	186.340	320.677
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	0.084916	0.084974	186.350	319.895
polished_wave3_3_raw_centered_shape_curve_aware_Fw	0.084910	0.084970	186.769	318.766
polished_wave3_3_curve_aware_pointwise_control_global	0.085063	0.085120	187.090	319.094
polished_wave4_1_smooth_l1_robust_loss_Fw	0.085061	0.085122	187.116	319.869
polished_periodic_temporal_convolution_Fw	0.085364	0.085443	187.236	316.322
polished_wave3_3_full_curve_composite_Fw	0.085191	0.085256	187.347	318.566
polished_wave4_3_mixture_density_k2_Fw	0.085189	0.085240	187.358	319.356
polished_wave5_1_harmonic_prior_pointwise_control_global	0.085374	0.085428	187.413	317.760
polished_wave3_3_full_curve_composite_global	0.085284	0.085341	187.596	317.599
polished_wave3_2_harmonic_residual_offset_global	0.085263	0.085320	187.765	318.467
polished_residual_harmonic_mlp_Fw	0.085646	0.085696	187.950	315.761
polished_periodic_mlp_harmonic_Fw	0.085830	0.085880	188.420	317.347
polished_wave5_1_harmonic_prior_smooth_l1_structured_Fw	0.086190	0.086245	188.491	322.579
polished_wave3_3_raw_centered_shape_curve_aware_global	0.085695	0.085750	188.662	318.403
polished_gru_sequence_global	0.086025	0.086083	189.005	317.427
polished_wave4_1_smooth_l1_robust_loss_global	0.086037	0.086090	189.280	317.088
polished_periodic_temporal_convolution_global	0.086168	0.086235	189.292	316.954
polished_temporal_convolution_global	0.086158	0.086217	189.546	315.846
polished_periodic_gru_sequence_Fw	0.086270	0.086329	189.713	317.966
polished_periodic_gru_sequence_global	0.086508	0.086565	190.171	321.113
polished_residual_harmonic_gru_sequence_sparse_rcim_global	0.086770	0.086811	190.773	318.810

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_feedforward_Fw	0.086931	0.087006	190.832	319.925
polished_wave4_1_log_cosh_robust_loss_global	0.086857	0.086909	190.908	322.502
polished_feedforward_global	0.087265	0.087342	191.515	323.389
polished_lstm_sequence_global	0.087177	0.087231	191.649	320.837
polished_gru_sequence_Fw	0.087172	0.087229	191.682	322.605
polished_residual_harmonic_lstm_sequence_dense240_Fw	0.087313	0.087440	192.137	319.266
polished_wave3_2_harmonic_residual_offset_Fw	0.087584	0.087638	192.572	321.302
polished_wave4_4_causal_tcn_latent_offset_residual_Fw	0.087696	0.087760	192.689	316.348
polished_wave3_3_curve_aware_pointwise_control_Fw	0.087578	0.087631	192.719	320.801
polished_periodic_lstm_sequence_Fw	0.087733	0.087808	192.894	319.177
polished_residual_harmonic_gru_sequence_dense240_Fw	0.087700	0.087819	192.937	323.637
polished_residual_harmonic_lstm_sequence_dense360_global	0.087693	0.088041	192.988	320.637
polished_wave3_2_clean_sequential_residual_offset_Fw	0.087791	0.087843	193.000	316.766
polished_wave4_3_mixture_density_k3_global	0.087895	0.087945	193.020	325.272
polished_residual_harmonic_gru_sequence_dense360_global	0.087694	0.088076	193.036	320.668
polished_residual_harmonic_mlp_global	0.088016	0.088064	193.130	317.519
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	0.087927	0.087967	193.386	321.315
polished_residual_harmonic_lstm_sequence_sparse_rcim_Fw	0.087932	0.087972	193.422	319.672
polished_lstm_sequence_Fw	0.088000	0.088052	193.497	320.948
polished_wave4_4_gru_latent_offset_residual_global	0.088207	0.088272	193.741	319.288
polished_residual_harmonic_gru_sequence_sparse_rcim_Fw	0.088164	0.088204	193.895	321.474
polished_wave4_4_causal_tcn_latent_offset_residual_global	0.088407	0.088466	194.038	321.840
polished_residual_harmonic_lstm_sequence_dense360_Fw	0.088280	0.088701	194.256	322.941
polished_wave3_2_clean_sequential_residual_offset_global	0.088402	0.088457	194.459	318.670
polished_periodic_lstm_sequence_global	0.088836	0.088902	194.827	324.051
polished_temporal_convolution_Fw	0.088705	0.088763	195.075	316.539
polished_residual_harmonic_lstm_sequence_dense240_global	0.088681	0.088799	195.120	322.794
polished_residual_harmonic_gru_sequence_dense360_Fw	0.088752	0.089230	195.144	323.795
polished_wave4_4_gru_latent_offset_residual_Fw	0.088796	0.088857	195.229	321.564
polished_wave3_1_sequential_residual_offset_probe_Fw	0.088949	0.088999	195.544	320.624
polished_wave3_1_sequential_residual_offset_probe_global	0.088966	0.089029	195.572	320.192
polished_periodic_mlp_global	0.089219	0.089281	195.795	320.557
polished_periodic_mlp_Fw	0.089150	0.089207	195.822	322.482
polished_residual_harmonic_gru_sequence_dense240_global	0.089108	0.089219	195.824	324.871
polished_periodic_mlp_harmonic_global	0.089445	0.089501	196.443	321.277

Polished RCIM Model-Bank Reproduction Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_rcim_model_bank_reproduction_XGBM19_Fw	0.032567	0.032674	71.319	126.434
polished_rcim_model_bank_reproduction_RF19_Fw	0.033927	0.034030	74.277	132.489
polished_rcim_model_bank_reproduction_GBM19_Fw	0.035161	0.035261	77.115	138.019
polished_rcim_model_bank_reproduction_HGBM19_Fw	0.035273	0.035387	77.409	139.268
polished_rcim_model_bank_reproduction_ET19_Fw	0.038726	0.038810	85.150	149.396
polished_rcim_model_bank_reproduction_DT19_Fw	0.039046	0.039124	85.908	150.919
polished_rcim_model_bank_reproduction_ERT19_Fw	0.039120	0.039206	86.046	149.564
polished_rcim_model_bank_reproduction_LGBM19_Fw	0.039198	0.039275	86.209	149.793
polished_rcim_model_bank_reproduction_SVM19_Fw	0.056787	0.056964	127.070	196.523
polished_rcim_model_bank_reproduction_MLP19_Fw	0.423630	0.521370	959.820	1956.574

Backward Comparison

Retuned Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_retuned_GBM19_Bw	0.002766	0.003300	5.398	12.280
rcim_retuned_ET19_Bw	0.003441	0.004028	7.021	15.287
rcim_retuned_ERT19_Bw	0.003551	0.004161	7.269	13.187
rcim_retuned_RF19_Bw	0.003649	0.004256	7.543	15.083
paper_retuned_best_Bw	0.003675	0.004284	7.572	15.645
rcim_retuned_SVM19_Bw	0.004016	0.004599	8.813	17.215
rcim_retuned_DT19_Bw	0.004578	0.005169	9.728	19.601
rcim_retuned_HGBM19_Bw	0.004683	0.005301	9.978	17.712
rcim_retuned_LGBM19_Bw	0.008105	0.008655	18.057	35.748
rcim_retuned_ELM19_Bw	0.008917	0.009518	20.169	51.896
rcim_retuned_XGBM19_Bw	0.010679	0.011209	24.184	48.082
rcim_retuned_MLP19_Bw	0.019115	0.023025	44.141	88.991

RCIM Model-Bank Reproduction Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
SVM19_Bw	0.004822	0.005116	10.864	25.533
track1_best_Bw	0.005027	0.005212	11.860	48.106
LGBM19_Bw	0.005037	0.005231	11.880	48.106
GBM19_Bw	0.005180	0.005363	12.252	49.984
DT19_Bw	0.005226	0.005409	12.359	48.860
ERT19_Bw	0.005258	0.005442	12.434	51.665
RF19_Bw	0.005392	0.005584	12.731	55.740
ET19_Bw	0.006273	0.006520	14.314	48.624
HGBM19_Bw	0.006619	0.006834	15.494	53.982
XGBM19_Bw	0.007991	0.008195	18.722	59.067
ELM19_Bw	0.010071	0.010486	23.034	52.159
MLP19_Bw	0.036681	0.044921	86.544	225.831

Wave 1 Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
tree_Bw	0.003258	0.003651	7.051	14.116
tree_global	0.003290	0.003702	7.118	13.703
periodic_mlp_Bw	0.003574	0.004006	7.807	13.970
periodic_mlp_global	0.003610	0.004030	7.879	13.661
residual_harmonic_mlp_global	0.003639	0.004044	7.967	14.489
feedforward_Bw	0.003644	0.004067	7.977	14.284
residual_harmonic_mlp_Bw	0.003672	0.004089	8.046	14.973
harmonic_regression_Bw	0.003745	0.004173	8.171	15.684
feedforward_global	0.003741	0.004188	8.186	15.259
harmonic_regression_global	0.018006	0.018276	41.275	83.603

Wave 2.1 Temporal Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
periodic_gru_sequence_global	0.002630	0.002876	6.010	12.693
periodic_lstm_sequence_Bw	0.002625	0.002877	6.013	15.382
periodic_lstm_sequence_global	0.002689	0.002956	6.098	14.674
lstm_sequence_global	0.003515	0.003944	7.666	12.502
lstm_sequence_Bw	0.003555	0.003985	7.767	14.507
gru_sequence_Bw	0.003626	0.004082	7.907	12.900
periodic_temporal_convolution_global	0.003604	0.003948	7.935	13.804
gru_sequence_global	0.003637	0.004080	7.946	14.580
periodic_temporal_convolution_Bw	0.003628	0.003987	7.979	13.839
temporal_convolution_Bw	0.003742	0.004166	8.184	13.908
temporal_convolution_global	0.003994	0.004438	8.798	18.339

Wave 2.3 Residual Harmonic Temporal Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_lstm_sequence_sparse_rcim_Bw	0.003440	0.003793	7.510	13.160
residual_harmonic_lstm_sequence_sparse_rcim_global	0.003491	0.003876	7.627	13.540
residual_harmonic_gru_sequence_sparse_rcim_Bw	0.003502	0.003857	7.654	12.829
residual_harmonic_gru_sequence_sparse_rcim_global	0.003584	0.003957	7.859	14.946
residual_harmonic_lstm_sequence_dense240_global	0.006591	0.008932	14.858	21.728
residual_harmonic_gru_sequence_dense240_global	0.006786	0.009221	15.282	20.990
residual_harmonic_lstm_sequence_dense240_Bw	0.007367	0.009945	16.660	22.200
residual_harmonic_gru_sequence_dense360_global	0.008181	0.011621	18.502	23.968
residual_harmonic_lstm_sequence_dense360_global	0.008860	0.013061	20.073	23.940
residual_harmonic_gru_sequence_dense240_Bw	0.008984	0.012987	20.358	26.370
residual_harmonic_gru_sequence_dense360_Bw	0.009370	0.013165	21.267	25.901
residual_harmonic_lstm_sequence_dense360_Bw	0.010268	0.014769	23.355	29.779

Global Model Direction Breakdown

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
feedforward_global	forward	0.003550	0.003979	7.916	16.195
feedforward_global	backward	0.003741	0.004188	8.186	15.259
feedforward_global	combined	0.003646	0.004083	8.051	15.583
gru_sequence_global	forward	0.003546	0.003975	7.869	14.344
gru_sequence_global	backward	0.003637	0.004080	7.946	14.580
gru_sequence_global	combined	0.003591	0.004028	7.907	14.389
harmonic_regression_global	forward	0.018314	0.018547	41.749	79.404
harmonic_regression_global	backward	0.018006	0.018276	41.275	83.603
harmonic_regression_global	combined	0.018160	0.018412	41.512	82.048
lstm_sequence_global	forward	0.003445	0.003863	7.642	12.032
lstm_sequence_global	backward	0.003515	0.003944	7.666	12.502
lstm_sequence_global	combined	0.003480	0.003903	7.654	12.430
periodic_gru_sequence_global	forward	0.002777	0.003023	6.267	13.580
periodic_gru_sequence_global	backward	0.002630	0.002876	6.010	12.693
periodic_gru_sequence_global	combined	0.002704	0.002949	6.139	13.200
periodic_lstm_sequence_global	forward	0.002726	0.002959	6.142	14.092
periodic_lstm_sequence_global	backward	0.002689	0.002956	6.098	14.674
periodic_lstm_sequence_global	combined	0.002707	0.002958	6.120	14.717
periodic_mlp_global	forward	0.003380	0.003791	7.493	14.349
periodic_mlp_global	backward	0.003610	0.004030	7.879	13.661
periodic_mlp_global	combined	0.003495	0.003910	7.686	14.284
periodic_temporal_convolution_global	forward	0.003407	0.003724	7.581	14.480
periodic_temporal_convolution_global	backward	0.003604	0.003948	7.935	13.804

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_temporal_convolution_global	combined	0.003506	0.003836	7.758	14.308
polished_feedforward_global	forward	0.087265	0.087342	191.515	323.389
polished_feedforward_global	combined	0.087265	0.087342	191.515	323.389
polished_gru_sequence_global	forward	0.086025	0.086083	189.005	317.427
polished_gru_sequence_global	combined	0.086025	0.086083	189.005	317.427
polished_harmonic_regression_global	forward	0.081589	0.081641	179.273	315.612
polished_harmonic_regression_global	combined	0.081589	0.081641	179.273	315.612
polished_lstm_sequence_global	forward	0.087177	0.087231	191.649	320.837
polished_lstm_sequence_global	combined	0.087177	0.087231	191.649	320.837
polished_periodic_gru_sequence_global	forward	0.086508	0.086565	190.171	321.113
polished_periodic_gru_sequence_global	combined	0.086508	0.086565	190.171	321.113
polished_periodic_lstm_sequence_global	forward	0.088836	0.088902	194.827	324.051
polished_periodic_lstm_sequence_global	combined	0.088836	0.088902	194.827	324.051
polished_periodic_mlp_global	forward	0.089219	0.089281	195.795	320.557
polished_periodic_mlp_global	combined	0.089219	0.089281	195.795	320.557
polished_periodic_mlp_harmonic_global	forward	0.089445	0.089501	196.443	321.277
polished_periodic_mlp_harmonic_global	combined	0.089445	0.089501	196.443	321.277
polished_periodic_temporal_convolution_global	forward	0.086168	0.086235	189.292	316.954
polished_periodic_temporal_convolution_global	combined	0.086168	0.086235	189.292	316.954
polished_residual_harmonic_gru_sequence_dense240_global	forward	0.089108	0.089219	195.824	324.871
polished_residual_harmonic_gru_sequence_dense240_global	combined	0.089108	0.089219	195.824	324.871
polished_residual_harmonic_gru_sequence_dense360_global	forward	0.087694	0.088076	193.036	320.668
polished_residual_harmonic_gru_sequence_dense360_global	combined	0.087694	0.088076	193.036	320.668
polished_residual_harmonic_gru_sequence_sparse_rcim_global	forward	0.086770	0.086811	190.773	318.810
polished_residual_harmonic_gru_sequence_sparse_rcim_global	combined	0.086770	0.086811	190.773	318.810
polished_residual_harmonic_lstm_sequence_dense240_global	forward	0.088681	0.088799	195.120	322.794
polished_residual_harmonic_lstm_sequence_dense240_global	combined	0.088681	0.088799	195.120	322.794
polished_residual_harmonic_lstm_sequence_dense360_global	forward	0.087693	0.088041	192.988	320.637
polished_residual_harmonic_lstm_sequence_dense360_global	combined	0.087693	0.088041	192.988	320.637
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	forward	0.087927	0.087967	193.386	321.315
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	combined	0.087927	0.087967	193.386	321.315
polished_residual_harmonic_mlp_global	forward	0.088016	0.088064	193.130	317.519
polished_residual_harmonic_mlp_global	combined	0.088016	0.088064	193.130	317.519
polished_temporal_convolution_global	forward	0.086158	0.086217	189.546	315.846
polished_temporal_convolution_global	combined	0.086158	0.086217	189.546	315.846
polished_tree_global	forward	0.066315	0.066379	145.420	257.447

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_tree_global	combined	0.066315	0.066379	145.420	257.447
polished_wave3_1_sequential_residual_offset_probe_global	forward	0.088966	0.089029	195.572	320.192
polished_wave3_1_sequential_residual_offset_probe_global	combined	0.088966	0.089029	195.572	320.192
polished_wave3_2_clean_sequential_residual_offset_global	forward	0.088402	0.088457	194.459	318.670
polished_wave3_2_clean_sequential_residual_offset_global	combined	0.088402	0.088457	194.459	318.670
polished_wave3_2_harmonic_residual_offset_global	forward	0.085263	0.085320	187.765	318.467
polished_wave3_2_harmonic_residual_offset_global	combined	0.085263	0.085320	187.765	318.467
polished_wave3_3_curve_aware_pointwise_control_global	forward	0.085063	0.085120	187.090	319.094
polished_wave3_3_curve_aware_pointwise_control_global	combined	0.085063	0.085120	187.090	319.094
polished_wave3_3_full_curve_composite_global	forward	0.085284	0.085341	187.596	317.599
polished_wave3_3_full_curve_composite_global	combined	0.085284	0.085341	187.596	317.599
polished_wave3_3_raw_centered_shape_curve_aware_global	forward	0.085695	0.085750	188.662	318.403
polished_wave3_3_raw_centered_shape_curve_aware_global	combined	0.085695	0.085750	188.662	318.403
polished_wave3_3_raw_offset_curve_aware_global	forward	0.084156	0.084216	185.372	318.550
polished_wave3_3_raw_offset_curve_aware_global	combined	0.084156	0.084216	185.372	318.550
polished_wave4_1_log_cosh_robust_loss_global	forward	0.086857	0.086909	190.908	322.502
polished_wave4_1_log_cosh_robust_loss_global	combined	0.086857	0.086909	190.908	322.502
polished_wave4_1_mae_robust_loss_global	forward	0.084350	0.084403	185.643	318.504
polished_wave4_1_mae_robust_loss_global	combined	0.084350	0.084403	185.643	318.504
polished_wave4_1_smooth_l1_robust_loss_global	forward	0.086037	0.086090	189.280	317.088
polished_wave4_1_smooth_l1_robust_loss_global	combined	0.086037	0.086090	189.280	317.088
polished_wave4_2_gaussian_nll_global	forward	0.082468	0.082526	181.304	319.127
polished_wave4_2_gaussian_nll_global	combined	0.082468	0.082526	181.304	319.127
polished_wave4_2_quantile_p10_p50_p90_global	forward	0.083629	0.083685	183.639	320.063
polished_wave4_2_quantile_p10_p50_p90_global	combined	0.083629	0.083685	183.639	320.063
polished_wave4_3_mixture_density_k2_global	forward	0.082547	0.082604	181.584	320.574
polished_wave4_3_mixture_density_k2_global	combined	0.082547	0.082604	181.584	320.574
polished_wave4_3_mixture_density_k3_global	forward	0.087895	0.087945	193.020	325.272
polished_wave4_3_mixture_density_k3_global	combined	0.087895	0.087945	193.020	325.272
polished_wave4_4_causal_tcn_latent_offset_residual_global	forward	0.088407	0.088466	194.038	321.840
polished_wave4_4_causal_tcn_latent_offset_residual_global	combined	0.088407	0.088466	194.038	321.840
polished_wave4_4_gru_latent_offset_residual_global	forward	0.088207	0.088272	193.741	319.288
polished_wave4_4_gru_latent_offset_residual_global	combined	0.088207	0.088272	193.741	319.288

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_wave5_1_harmonic_prior_pointwise_control_global	forward	0.085374	0.085428	187.413	317.760
polished_wave5_1_harmonic_prior_pointwise_control_global	combined	0.085374	0.085428	187.413	317.760
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	forward	0.084916	0.084974	186.350	319.895
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	combined	0.084916	0.084974	186.350	319.895
residual_harmonic_gru_sequence_dense240_global	forward	0.006534	0.008959	14.732	20.898
residual_harmonic_gru_sequence_dense240_global	backward	0.006786	0.009221	15.282	20.990
residual_harmonic_gru_sequence_dense240_global	combined	0.006660	0.009090	15.007	20.940
residual_harmonic_gru_sequence_dense360_global	forward	0.007844	0.011211	17.679	22.520
residual_harmonic_gru_sequence_dense360_global	backward	0.008181	0.011621	18.502	23.968
residual_harmonic_gru_sequence_dense360_global	combined	0.008012	0.011416	18.090	23.559
residual_harmonic_gru_sequence_sparse_rcim_global	forward	0.003287	0.003611	7.282	12.461
residual_harmonic_gru_sequence_sparse_rcim_global	backward	0.003584	0.003957	7.859	14.946
residual_harmonic_gru_sequence_sparse_rcim_global	combined	0.003435	0.003784	7.571	13.869
residual_harmonic_lstm_sequence_dense240_global	forward	0.006247	0.008597	14.061	20.066
residual_harmonic_lstm_sequence_dense240_global	backward	0.006591	0.008932	14.858	21.728
residual_harmonic_lstm_sequence_dense240_global	combined	0.006419	0.008765	14.460	20.628
residual_harmonic_lstm_sequence_dense360_global	forward	0.008760	0.012992	19.759	24.068
residual_harmonic_lstm_sequence_dense360_global	backward	0.008860	0.013061	20.073	23.940
residual_harmonic_lstm_sequence_dense360_global	combined	0.008810	0.013026	19.916	24.052
residual_harmonic_lstm_sequence_sparse_rcim_global	forward	0.003245	0.003562	7.192	12.555
residual_harmonic_lstm_sequence_sparse_rcim_global	backward	0.003491	0.003876	7.627	13.540
residual_harmonic_lstm_sequence_sparse_rcim_global	combined	0.003368	0.003719	7.409	13.041
residual_harmonic_mlp_global	forward	0.003649	0.004051	8.123	14.713
residual_harmonic_mlp_global	backward	0.003639	0.004044	7.967	14.489
residual_harmonic_mlp_global	combined	0.003644	0.004048	8.045	14.717
sequential_residual_offset_probe_global	forward	0.003425	0.003839	7.599	12.346
sequential_residual_offset_probe_global	combined	0.003425	0.003839	7.599	12.346
temporal_convolution_global	forward	0.003508	0.003928	7.792	14.045
temporal_convolution_global	backward	0.003994	0.004438	8.798	18.339
temporal_convolution_global	combined	0.003751	0.004183	8.295	17.247
track2f_bis_clean_sequential_residual_offset_global	forward	0.003420	0.003841	7.590	13.476
track2f_bis_clean_sequential_residual_offset_global	combined	0.003420	0.003841	7.590	13.476
track2f_bis_harmonic_residual_offset_global	forward	0.003255	0.003547	7.224	12.114
track2f_bis_harmonic_residual_offset_global	combined	0.003255	0.003547	7.224	12.114
track2g_curve_aware_full_curve_composite_global	forward	0.085677	0.085733	188.514	319.710

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_full_curve_composite_global	combined	0.085677	0.085733	188.514	319.710
track2g_curve_aware_pointwise_control_global	forward	0.003499	0.003784	7.771	13.513
track2g_curve_aware_pointwise_control_global	combined	0.003499	0.003784	7.771	13.513
track2g_curve_aware_raw_centered_shape_global	forward	0.003248	0.003556	7.218	12.268
track2g_curve_aware_raw_centered_shape_global	combined	0.003248	0.003556	7.218	12.268
track2g_curve_aware_raw_offset_global	forward	0.003359	0.003630	7.429	12.737
track2g_curve_aware_raw_offset_global	combined	0.003359	0.003630	7.429	12.737
track2h_gaussian_nll_global	forward	0.002951	0.003246	6.524	15.237
track2h_gaussian_nll_global	combined	0.002951	0.003246	6.524	15.237
track2h_l_causal_tcn_offset_residual_global	forward	0.003267	0.003715	7.241	12.649
track2h_l_causal_tcn_offset_residual_global	combined	0.003267	0.003715	7.241	12.649
track2h_l_gru_offset_residual_global	forward	0.087641	0.087704	192.524	320.620
track2h_l_gru_offset_residual_global	combined	0.087641	0.087704	192.524	320.620
track2h_log_cosh_robust_global	forward	0.003515	0.003825	7.792	14.729
track2h_log_cosh_robust_global	combined	0.003515	0.003825	7.792	14.729
track2h_mae_robust_global	forward	0.003235	0.003544	7.189	12.856
track2h_mae_robust_global	combined	0.003235	0.003544	7.189	12.856
track2h_mdn_k2_global	forward	0.003263	0.003582	7.261	14.354
track2h_mdn_k2_global	combined	0.003263	0.003582	7.261	14.354
track2h_mdn_k3_global	forward	0.003415	0.003713	7.594	15.012
track2h_mdn_k3_global	combined	0.003415	0.003713	7.594	15.012
track2h_quantile_p10_p50_p90_global	forward	0.003188	0.003469	7.059	12.765
track2h_quantile_p10_p50_p90_global	combined	0.003188	0.003469	7.059	12.765
track2h_smooth_l1_robust_global	forward	0.003222	0.003507	7.161	12.021
track2h_smooth_l1_robust_global	combined	0.003222	0.003507	7.161	12.021
tree_global	forward	0.002998	0.003364	6.590	11.601
tree_global	backward	0.003290	0.003702	7.118	13.703
tree_global	combined	0.003144	0.003533	6.854	13.314
wave3_harmonic_prior_residual_pointwise_control_global	forward	0.003279	0.003550	7.289	14.109
wave3_harmonic_prior_residual_pointwise_control_global	combined	0.003279	0.003550	7.289	14.109
wave3_harmonic_prior_residual_smooth_l1_structured_global	forward	0.087837	0.087893	192.823	321.880
wave3_harmonic_prior_residual_smooth_l1_structured_global	combined	0.087837	0.087893	192.823	321.880
wave52b_offset_centered_shape_harmonic_global	forward	0.080306	0.080373	176.296	312.921
wave52b_offset_centered_shape_harmonic_global	combined	0.080306	0.080373	176.296	312.921

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-05-08-06-21__track2_full_directional_family_matrix_track2_dataset_surface_simplified_dataset_forward_2026_07_04\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-05-08-06-21__track2_full_directional_family_matrix_track2_dataset_surface_simplified_dataset_forward_2026_07_04\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional paper-reference, Wave 1, and Wave 2.1 models are never evaluated on the opposite direction. Global Wave models remain valid on both directions and are therefore shown in the directional sections and again in the global breakdown. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.